

Distributed LLM Training over Passive Optical Networks: Performance Impact of Dynamic Bandwidth Allocation

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Abstract We implement data-parallel LLM training on a two-node 25GS-PON testbed and evaluate AllReduce performance under varying delay tolerance and bandwidth configurations. Results show that higher delay tolerance increases communication latency, demonstrating that properly configured PONs can feasibly support small- and medium-scale distributed training workloads. ©2026 The Author(s)

Introduction

Conventional switch-based networks offer high flexibility and bandwidth for large-scale AI training but face significant scalability, energy, and cost challenges^[1]. Passive optical networks (PONs) present a compelling alternative for edge-oriented and access-network-based distributed AI training, where compute nodes are typically medium- to low-end servers or intelligent devices deployed close to data sources. By replacing power-hungry active switching fabrics with passive optical splitters, PON architectures can substantially reduce network power consumption, capital cost, and system complexity^[2]. Moreover, the inherent data locality of PON-connected nodes enables training paradigms where data remains at the edge, reducing reliance on centralized storage, lowering backhaul traffic, and enhancing data privacy—an important requirement for Internet of Things (IoT)^[3], federated learning^{[4],[5]}, and edge intelligence applications^[6]. The widespread deployment of PON infrastructure and its steadily increasing bandwidth capacity make it a promising platform for supporting distributed edge workloads, including collaborative and incremental model training across access and aggregation networks^{[7],[8]}. Dynamic bandwidth allocation (DBA) is essential for efficiently supporting such workloads over shared optical networks. DBA de-

sign must balance fundamental trade-offs among latency, efficiency, fairness, and scalability to support distributed training over PON-based edge infrastructures^{[9],[10]}.

In this paper, we experimentally study a transformer-based model trained with data parallelism over a symmetric 25GS-PON^[11], focusing on the impact of DBA parameters on bursty, communication-intensive training traffic on the upstream direction. By varying delay tolerance (DT), which limits the maximum interval between bursts, and bandwidth configurations, we measure their effects on AllReduce and computation performance.

Experiment

Fig. 1(a) illustrates two-node data-parallel training. The training dataset is distributed across nodes, and the model is replicated on each node. In each step, nodes process local mini-batches to compute forward and backward passes, synchronize gradients via an AllReduce operation, and then update identical model parameters before proceeding to the next step. The AllReduce communication generates bursty, periodic traffic. Fig. 1(b)(i) illustrates traffic provided with sufficient bandwidth and a short grant cycle, which together achieve low transmission latency. However, in practice, longer grant cycles or insufficient bandwidth are common due to shared

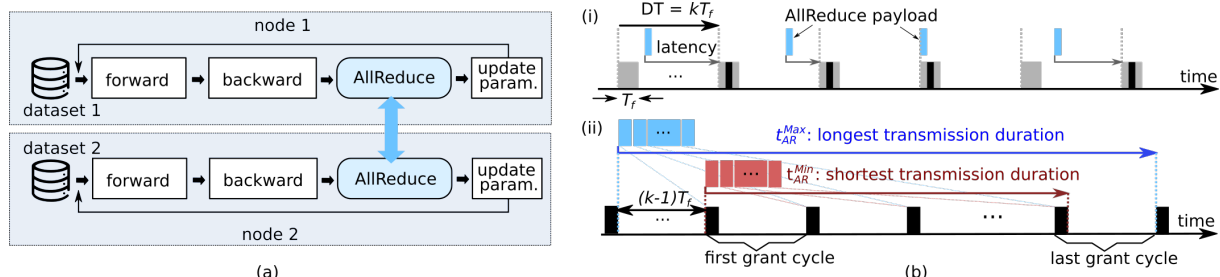


Fig. 1: (a) Two-node data-parallel training with gradient synchronization via AllReduce; (b) DBA provisioning with sufficient bandwidth and frequent grants (i), and with limited bandwidth and infrequent grants for a large message (ii).

network resources and multiple competing optical network units (ONUs), increasing overall delay. Besides, the size of AllReduce messages—determined by the number of model parameters and data precision—is typically large. In a PON, these messages are divided into multiple chunks for transmission.

Fig. 1(b)(ii) illustrates the transmission duration range of a large message under a given DT, defined as kT_f , where k is the number of tolerable PON frames and T_f is the 125- μ s frame duration. The blue solid line represents the longest AllReduce transmission duration, t_{AR}^{Max} , which occurs when the transmission experiences the maximum waiting time of $(k-1)T_f$ before the first effective grant is received. The transmission duration is given by $t_{AR}^{Max} = (k-1)T_f + \lceil \frac{P}{C} \rceil kT_f$, where P is the AllReduce data payload, C is the assigned capacity per grant and $\lceil \cdot \rceil$ represents ceiling function. The red solid line represents the shortest transmission duration, t_{AR}^{Min} , which occurs when the ONU can transmit immediately upon receiving the first grant with negligible waiting time. It can be expressed as: $t_{AR}^{Min} = \lfloor \frac{P}{C} \rfloor kT_f + T_f$, where $\lfloor \cdot \rfloor$ represents floor function. The duration difference Δt , defined as the difference between t_{AR}^{Max} and t_{AR}^{Min} , can be calculated as: $\Delta t = 2(k-1)T_f$. This indicates that the range of t_{AR} per training step increases as the DT value increases. Protocol overhead and device scheduling delays are not included in the above formula; therefore, the measured t_{AR} is slightly larger.

Fig. 2(a) illustrates the experimental setup. The testbed is built on a 25GS-PON in which one server is connected to a 100G port of an opti-

cal line terminal (OLT) through 100G QSFP pluggables, and the other server is connected to one ONU through 25G SFP pluggables. Both servers are equipped with Nvidia RTX A6000 Ampere GPUs. During the measurements, the OLT and ONU are linked over standard single-mode fibers (SSMFs) with different lengths up to 20 km. Built on this PON infrastructure, the two servers perform data-parallel training, each holding a separate data partition. They compute local gradients and synchronize them through the OLT.

We used a small but efficient pre-trained transformer model, Femto^[12] with 0.243M parameters, and fine-tuned it on the IMDB dataset^[13]—a large collection of movie reviews—for a binary classification task. The training was performed using FP16 precision. The PON was configured with fixed information rates of 100 Mbit/s, 200 Mbit/s, 500 Mbit/s, and 1 Gbit/s. The DT value starts at 250 μ s and doubles up to the maximum effective limit for each bandwidth, beyond which the number of data blocks per grant remains unchanged. We collected the training duration over 100 epochs, during which all configurations achieved their highest evaluation accuracy (77.0%-78.3% between epochs 38 and 71), see Fig. 2(b), which illustrates the accuracy trend at 100 Mbit/s.

Result

Fig. 2(c) presents the histogram of t_{AR} under different DT at the bandwidth of 100 Mbit/s. The duration range for DT=4 ms is approximately twice that for DT=2 ms, and that for DT=2 ms is approximately twice that for DT=1 ms, confirming the

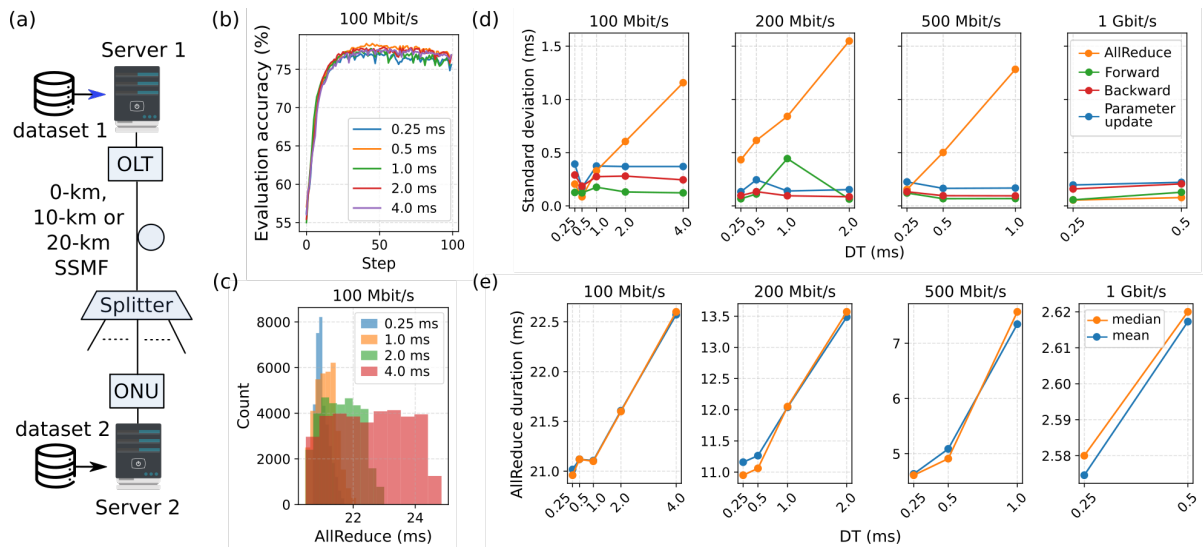


Fig. 2: (a) Experimental setup, (b) evaluation accuracy over 100 epochs and (c) histogram of AllReduce duration t_{AR} at 100 Mbit/s under different delay tolerance (DT) settings. (d) Standard deviation of AllReduce, forward pass, backward pass, and model parameter update durations, and (e) mean and median t_{AR} per training step under bandwidths of 100 Mbit/s, 200 Mbit/s, 500 Mbit/s, and 1 Gbit/s with varying DT values.

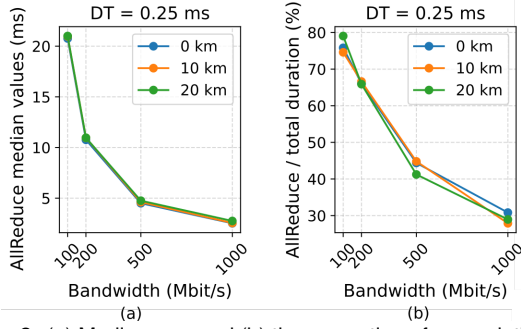


Fig. 3: (a) Median t_{AR} and (b) the proportion of t_{AR} relative to the total communication and computation time across different bandwidths and transmission, with DT of 0.25 ms.

proportional relationship with Δt . Fig. 2(d) shows the standard deviation of t_{AR} alongside those of the computation processes—forward pass, backward pass, and parameter update—for all configurations. Consistent with the increasing range of t_{AR} at higher DT values, its standard deviation rises noticeably under bandwidths of 100 Mbit/s, 200 Mbit/s, and 500 Mbit/s. In contrast, the standard deviations of the forward pass, backward pass, and parameter update remain relatively stable, as these computations are primarily determined by local hardware performance and minimally affected by network conditions. For 1 Gbit/s, the standard deviation increases less noticeably, likely because the fast t_{AR} leads to minimal variation. Fig. 2(e) shows the mean and median t_{AR} for different DT values under varying bandwidths. As bandwidth increases from 100 Mbit/s to 1 Gbit/s, the t_{AR} decreases from 21.0–22.6 ms to around 2.6 ms, confirming that higher network bandwidth enables faster inter-node communication. Meanwhile, for all bandwidth levels, both the mean and median t_{AR} durations increase (for 200 Mbit/s, 500 Mbit/s, and 1 Gbit/s) or show an upward trend (for 100 Mbit/s) as DT becomes larger. We evaluated the impact of node distances (0 km, 10 km, and 20 km) on AllReduce communication and found that distance has a minimal effect. Both the absolute t_{AR} durations and the ratio of t_{AR} to total step duration remain very similar across all distances and bandwidths, see Fig. 3.

Intelligent Bandwidth Assignment

When gradient computation during the backward pass, t_{bwd} , and gradient synchronization via AllReduce, t_{AR} , proceed concurrently with nearly equal durations, communication can be largely hidden by computation^{[14],[15]}. Fig. 4(a) shows that the fitted curve of the mean t_{AR} intersects the linear fit of mean t_{bwd} near 1.5 Gbit/s for the model Femto. When the assigned bandwidth is below 1.5 Gbit/s, $t_{AR} > t_{bwd}$, causing

the GPU to remain idle while waiting for AllReduce to complete. When the assigned bandwidth is above 1.5 Gbit/s, $t_{AR} < t_{bwd}$, resulting in underutilization of the allocated bandwidth. Therefore, assigning bandwidth that maximizes communication-computation overlap is critical to sustaining training efficiency while ensuring fair bandwidth sharing among coexisting PON users.

The ITU-T supplementary document for a future Very High Speed PON (VHSP) standard targeting 200 Gbit/s^[16] greatly enhances the feasibility of distributed AI training over PONs. Under such high-speed access networks, small-scale models such as BERT-base (110M) and BERT-large (340M) can reduce t_{AR}^{Min} to below 0.3 s at 4 Gbit/s and 10 Gbit/s, respectively. For medium-scale LLMs, GPT-3 XL (1.3B), LLaMA-2 7B, and 13B achieve t_{AR}^{Min} below 0.55 s at 20 Gbit/s, 110 Gbit/s, and 200 Gbit/s, respectively. The backward pass time can be expressed as $t_{bwd} \approx \frac{4MS}{GPU_{FLOPS}}$, where M is the model parameter count, S denotes tokens processed per step, and GPU_{FLOPS} represents the effective floating-point throughput of the GPU. Thus, for a fixed DT value in medium-scale LLM settings, the required bandwidth for maximum communication-computation overlap scales approximately linearly with GPU compute performance: $B \propto GPU_{FLOPS}$. We estimate computation latency using workstation-class RTX 6000 GPUs spanning Ampere (300 TFLOPS), Ada (1.4 PFLOPS), and Blackwell (2.2 PFLOPS) architectures. Fig 4(b) shows that, under our testbed settings, near-optimal balance can be achieved with assigned bandwidths of approximately 16 Gbit/s, 75 Gbit/s, and 115 Gbit/s for GPT-3 XL (1.3B), Llama-2 7B and 13B, processed with various GPUs. These findings highlight the necessity of model- and hardware-aware intelligent DBA mechanisms in next-generation PONs to efficiently support emerging AI training workloads.

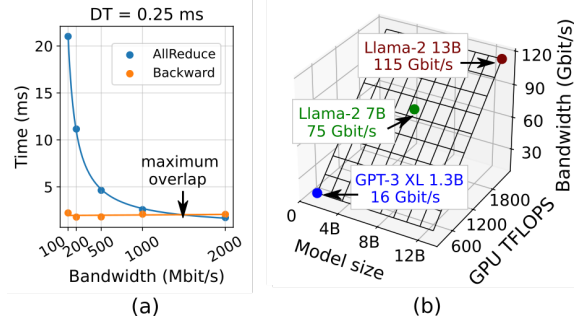


Fig. 4: (a) Measured mean t_{AR} and mean t_{bwd} under different bandwidths with DT=0.25 ms and node distance of 0 km. (b) 3D plot of model size, GPU compute, and required bandwidth for maximum communication-computation overlap.

References

- [1] S. Rumley, D. Nikolova, R. Hendry, Q. Li, D. Calhoun, and K. Bergman, "Silicon photonics for exascale systems", *Journal of Lightwave Technology*, vol. 33, no. 3, pp. 547–562, 2015.
- [2] G. Kramer, B. Mukherjee, and A. Maislos, "Ethernet passive optical networks", *IP over WDM: Building the Next-Generation Optical Internet*, pp. 229–275, 2003.
- [3] X. Lim *et al.*, "A survey of federated learning for edge computing: Research problems and solutions", *High-Confidence Computing*, 2021.
- [4] J. Li, L. Chen, and J. Chen, "Scalable federated learning over passive optical networks", in *Optical Fiber Communication Conference (OFC)*, Optica Publishing Group, 2021, W6A–36.
- [5] O. J. Ciceri, C. A. Astudillo, Z. Zhu, and N. L. da Fonseca, "Federated learning over next-generation Ethernet passive optical networks", *IEEE network*, vol. 37, no. 1, pp. 70–76, 2022.
- [6] D. Xu, T. Li, Y. Li, *et al.*, "Edge intelligence: Architectures, challenges, and applications", *arXiv*, 2020.
- [7] F. Zhu, T. Zeng, X. Huang, M. Jiang, and D. Zhang, "Field demonstration of lossless RDMA over 50G-PON", in *2025 Asia Communications and Photonics Conference (ACP)*, IEEE, 2025, pp. 1–5.
- [8] S. Bidkar, A. Gudal, L. Drabeck, J. Simsarian, and M. Khan, "AI agent-driven network-aware decentralized compute resource brokering", in *Optical Fiber Communications Conference (OFC)*, 2026, M3Z.15.
- [9] K. Christodoulopoulos, S. Bidkar, W. Lautenschlaeger, T. Pfeiffer, and R. Bonk, "Deterministic passive optical networking", *Journal of Optical Communications and Networking*, vol. 16, no. 2, pp. 180–192, 2024.
- [10] J. Maes, S. Bidkar, M. Straub, T. Pfeiffer, and R. Bonk, "Efficient transport of enhanced CPRI fronthaul over PON", *Journal of Optical Communications and Networking*, vol. 16, no. 2, A136–A142, 2024.
- [11] 25GS-PON MSA, <https://www.25gspon-msa.org/>, Accessed: 2026-04-01.
- [12] NeuML, *Neuml/bert-hash-femto*, Accessed: 2026-02-08; Apache-2.0 license, 2025. [Online]. Available: <https://huggingface.co/NeuML/bert-hash-femto>.
- [13] K. Enevoldsen, I. Chung, I. Kerboua, *et al.*, "MMTEB: Massive multilingual text embedding benchmark", *arXiv preprint arXiv:2502.13595*, 2025. DOI: 10.48550/arXiv.2502.13595. [Online]. Available: <https://arxiv.org/abs/2502.13595>.
- [14] M. Shoenybi, M. Patwary, R. Puri, P. LeGresley, J. Casper, and B. Catanzaro, "Megatron-LM: Training multi-billion parameter language models using model parallelism", *arXiv preprint arXiv:1909.08053*, 2019.
- [15] J. Rasley, S. Rajbhandari, O. Ruwase, and Y. He, "Deepspeed: System optimizations enable training deep learning models with over 100 billion parameters", in *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery Data Mining*, 2020.
- [16] ITU-T, *Point to multipoint passive optical access system requirements and transmission technologies above 50 Gbit/s per wavelength*, <https://www.itu.int/rec/T-REC-G.Supp88/en>, Supplement 88 to ITU-T G series Recommendations; In force (10/2025), 2025.